Predicting TCR Cross-Reactivity Before It Reaches a Patient

Sergio E. Mares, PhD · Adimab, Mountain View, CA · Center for Computational Biology, UC Berkeley · Department of Neurosurgery, UCSF

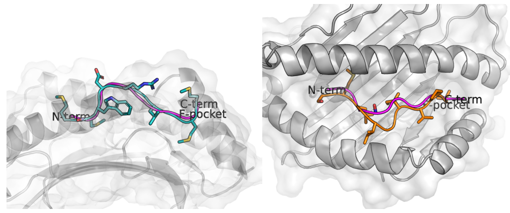

Figure 1: **Proposed framework.** A two-state discrimination test for structure-based peptide design in a cross-reactive pMHC–TCR system. Diffusion models sample from the cognate conformation (teal, left) toward a known off-target conformation (orange, right), two distinct peptides the same receptor engages. Each generated backbone (magenta) is scored against both, and credited only if substantially closer to the one it is claimed to match.

1 Motivation

Engineered T-cell receptor (TCR) therapies have entered clinical practice with the 2024 accelerated approval of afamitresgene autoleucel, the first TCR gene therapy and first engineered cell therapy for a solid tumor (Singh 2024). Their extension to personalized medicine is constrained by off-target cross-reactivity, which remains difficult to anticipate preclinically. The limitation is not theoretical: an affinity-enhanced MAGE-A3 receptor that had passed extensive preclinical characterization produced fatal cardiogenic shock in the first two treated patients (Linette et al. 2013), and the causative epitope, a titin-derived self-peptide sharing five of nine residues with the intended target, was resolved only retrospectively (Cameron et al. 2013).

Three factors place this liability beyond current methods. Experimental validation scales poorly against a presented peptide space of $\sim 10^{11}$ sequences, compounded by the narrow clinical window in aggressive solid tumors. HLA polymorphism leaves binding data sparse across most alleles, and disproportionately so for underrepresented populations, precisely where personalized therapy is most needed. Positional scanning mutagenesis characterizes one receptor at a time and does not scale to the combinatorics of peptide–TCR interaction.

2 Proposed Framework

I propose to address this with two complementary tools of my own design: **ESMCBA**, a binding-affinity model using protein language models with domain adaptation that recovers signal for data-scarce HLA alleles (Mares, Espinoza, and Ioannidis 2025a), and a **structure-guided generative framework** using diffusion models conditioned on the MHC groove to sample compatible peptides across high-priority alleles (Mares, Espinoza, and Ioannidis 2025b). Together they permit computational screening of a receptor's cross-reactive neighborhood before clinical exposure.

Such a screen requires an evaluation standard the field lacks. Generative backbone models are scored by deviation to a single crystal structure, which cannot separate a design genuinely matching a target conformation from one merely proximal to the groove; sequence models are summarized by one average recovery rate, obscuring which constraints produce the signal. I propose each claim instead be evaluated by discrimination between two structurally defined alternatives (Fig. 1).

Aim 1: Backbone conformation. Cross-reactive TCRs engage chemically distinct peptides through distinct bound conformations, a two-state system in which off-target recognition is structurally explicit. We will assess whether a generative model, given only receptor-side contacts and no peptide template, can be steered to a target conformation, including deliberately to the conformation *other* than that nearest its conditioning input. Thresholds will be calibrated against native molecular-dynamics ensembles rather than a single snapshot, with independent criteria (proximity, anchor identity, burial depth) required to agree.

Aim 2: Sequence constraints. Inverse-folding models recover native TCR interface residues well above chance (Ribeiro-Filho et al. 2024), though on the receptor side. We will apply them to the *peptide* across solved pMHC–TCR structures, with the TCR present and computationally removed, to identify which positions a receptor reads. These define the axes along which a cross-reactive partner may differ, and therefore where a safety screen must search.

3 Expected Outcomes

This fellowship would develop that screen into a routine preclinical safeguard, converting cross-reactivity from an unanticipated failure mode into a predictable, testable step.

References

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